QIXIN WANG

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EDUCATION

National University of Singapore — **Singapore**
Doctor of Philosophy - Electrical & Computer Engineering — 2026-2030

Tsinghua University — **Beijing, China**
Master of Engineering - Computer Science and Technology — 2022 - 2025
Overall GPA: 3.91/4.0

Beihang University — **Beijing, China**
Bachelor of Fine Arts - Digital Media Art — 2017 - 2021
Overall GPA: 3.61/4.0

RESEARCH INTEREST

Human-Computer Interaction, Technology-Enhanced Learning, Learning Analytics, Multimodal Learning Interaction, Accessibility, Human-Centered AI, Data-Driven Education

PUBLICATION

- **Qixin Wang**, Songtao Zhou, Zeyu Jin, Shikun Sun, Qinghuan Lan, Chenglin Guo, Xiaoyu Qin. 2025. *Research on Vision Accessible System for Blind and Low-vision People*. Accepted by the Meeting of National Conference on Man-Machine Speech Communication **(NCMMSC 2025)** (Multimodal, HCI, Accessibility)
- Songtao Zhou, Xiaoyu Qin, Yixuan Zhou, **Qixin Wang,** Zeyu Jin, Zixuan Wang, Zhiyong Wu, Jia Jia. 2025 *HarmoniVox: Painting Voices to Match the Avatar's Soul.* To appear in Proceedings of the 33rd ACM International Conference on Multimedia **(ACM MM '25)**. (Human-AI Interaction, Avatar)
- **Qixin Wang**, Songtao Zhou, Zeyu Jin, Chenglin Guo, Shikun Sun, Xiaoyu Qin. 2025. *V-CASS: Vision-context-aware Expressive Speech Synthesis for Enhancing User Understanding of Videos*. To appear in Proceedings of the 2025 International Joint Conference on Neural Networks **(IJCNN 2025)**. (Multimodal Speech Synthesis)
- **Qixin Wang**, Chenglin Guo, Songtao Zhou, and Xiaoyu Qin. 2024. *Emotional Speech Synthesis via Visual Context Perception*. Accepted and oral presented at the Meeting of National Conference on Man-Machine Speech Communication **(NCMMSC 2024)**. (Wellbeing, Multimodal Perception)
- Zeyu Jin, Jia Jia, **Qixin Wang**, Kehan Li, Shuoyi Zhou, Songtao Zhou, Xiaoyu Qin, and Zhiyong Wu. 2024. *SpeechCraft: A Fine-Grained Expressive Speech Dataset with Natural Language Description*. In Proceedings of the 32nd ACM International Conference on Multimedia **(ACM MM '24)**. (Multimodal Speech Synthesis)
- Xiaohan Li, **Qixin Wang**, Zishan Wang, Zeyu Jin, and Jia Jia. 2024. *SoulSkipper: A Voice-Controlled Emotional Adaptive Game to Complement Therapy for Social Anxiety Disorder*. In Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems **(CHI EA '24)**. (Gamification, Mental Wellbeing)
- Zeyu Jin, Zixuan Wang, **Qixin Wang**, Jia Jia, Ye Bai, Yi Zhao, Hao Li, and Xiaorui Wang. 2023. *HoloSinger: Semantics and Music Driven Motion Generation with Octahedral Holographic Projection*. In Proceedings of the 31st ACM International Conference on Multimedia **(ACM MM '23)**. (Mixed Reality)
- Jiao, Y., **Wang, Q.**, Xu, Y. (2023). *Graphical Tactile Display Application: Design of Digital Braille Textbook and Initial Findings*. In: Wang, D., *et al.* Haptic Interaction. **(AsiaHaptics 2022)**. (Haptics, Accessibility, HCI)
- Yang Jiao, **Qixin Wang**, Yingqing Xu. *Research on Haptic Graphic Displays: Digitized Braille Textbook Design*. Accepted and Poster presented at the track of the CHCI in the 18th Conference of Harmonious Human-Machine Environment Meeting **(HHME-CHCI 2022)**. (Haptics, Accessibility, HCI)

RESEARCH EXPERIENCE

Research on Vision Accessible System for Blind and Low-vision People (Master's Thesis Project)

Department of Computer Science and Technology, Tsinghua University, 2024 - 2025

- Designed and implemented a multimodal agent-based accessibility system enabling blind and low-vision (BLV) users to independently watch short videos through expressive and adaptive audio description, and active exploration with the video content through voice and tactile interaction.
- Led the end-to-end research cycle, including requirement interviews, qualitative and quantitative analysis, system design, prototyping, and iterative improvement.
- Applied thematic analysis on user interview and focus group transcripts to identify recurring accessibility needs and pain points, and refine interaction strategies.

- Conducted mixed-methods evaluation with BLV participants, applying SUS, NASA-TLX, and comprehension testing, achieving increase in comprehension compared to baseline systems.
- Outcome: **First author** paper accepted by **NCMMSC 2025**.

Multimodal Expressive Speech Synthesis: Models, Datasets, and Applications
Department of Computer Science and Technology, Tsinghua University, 2023 - 2025

- **V-CASS: Vision-Context-Aware Expressive Speech Synthesis**
  - Proposed a vision-context-aware approach to enhance user understanding of videos.
  - Led system design and evaluation;
  - **First-author** paper accepted by **IJCNN 2025**. Demo page available at https://v-cass.github.io/V-CASS/
- **Emotional Speech Synthesis via Visual Context**
  - Investigated how visual context improves emotional expressiveness in synthesized speech.
  - Designed user studies evaluating comprehension and expressiveness.
  - **First-author** paper oral presented at **NCMMSC 2024**.
- **SpeechCraft: A Fine-Grained Expressive Speech Dataset with Natural Language Description**
  - Collaborated on the **open-source** dataset *SpeechCraft*, responsible for data collection and the design/implementation of the data processing pipeline.
  - Co-author paper accepted at **ACM MM'24**, open-source dataset available at https://github.com/thuhcsi/SpeechCraft.
- **HarmoniVox: Avatar-Aligned Voice Synthesis**
  - Explored how speech synthesis can reflect and align with avatar identities.
  - Responsible for model testing, designing and conducting user experiments, and figure design/visualization.
  - Co-author paper to appear at **ACM Multimedia 2025**. Demo page available at https://harmonivox.github.io/harmonivox/

SoulSkipper: Voice-Controlled Emotional Adaptive Game for Social Anxiety Therapy
Department of Computer Science and Technology, Tsinghua University, 2023 - 2024

- This research designed and prototyped a serious game that leverages **voice input** as both a control channel and an indicator of emotional state, supporting CBT-based behavioral training for social anxiety.
- Integrated adaptive interaction mechanisms that respond to vocal patterns, aiming to enhance engagement and therapeutic outcomes.
- Co-author paper accepted at **CHI 2024.**

Dream-Visualization Art Community Project
The Future Laboratory, Tsinghua University, 2022 - 2022

- This project translated REM sleep and EEG signals into generative visual art elements, developing an exploratory tool for stress relief, cognitive enhancement, health management, and sleep monitoring.
- Led the design and maintenance of the interactive visualization program and community website; independently completed graphic design tasks.
- Exhibited at the **Asia Digital Art Exhibition** and **Taobao Maker Festival**.

Graphical Tactile Display for Digital Braille Textbooks
The Future Laboratory, Tsinghua University, 2021 - 2022

- This project aimed to build a tactile display allowing BLV users to access both text and graphics via a dynamic dot matrix interface.
- Participated in mechanical design improvements, test program development, and prototype assembly. Explored the principle and method of digitalized braille textbook, analyzed findings to improve accessibility in educational contexts.
- Co-author papers publicated at **AsiaHaptics 2022** and **HHME-CHCI 2022**.

GRANTS & FELLOWSHIP & AWARDS

- Second-Class Scholarship of Tsinghua University — Awarded Fall Semester 2024
- Second-Class Scholarship of Tsinghua University — Awarded Fall Semester 2023
- Best Paper Award-Finalist in Asia Haptics 2022 — Nov. 2022
- Best Paper Award-Finalist in HHME-CHCI 2022 — Sept. 2022
- Outstanding Graduate Awards of Beihang University — Awarded Jun. 2021
- Second-Class Scholarship of Beihang University — Awarded Fall Semester 2018

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- Excellent Paper Presentation in the 4th HEED International Collegiate Conference of China May.2018

SKILLS

- **User-centered design method in research**:
  - User studies (interviews, focus groups, usability testing, SUS, NASA-TLX, A/B testing, thematic analysis).
  - Prototyping and evaluation methods (personas, journey mapping, heuristic evaluation, card sorting).
- **Data & Tools:** Python (Data Processing, Automatic Workflow), HTML/CSS/JS (Web front-end), C# (basic), R (basic), Unity, Arduino & Processing, Touch Designer.
- **Design Ability:** IxD, Graphic & Web design, VR & AR prototyping.
- **Communication:**
  - Multilingual (English, Mandarin, Japanese, Korean).
  - Experience collaborating across interaction design, computer science, music, psychology and architecture.
- **Portfolio (Interaction and UX Design):** Available at https://sqxw7n.github.io/mypage/#projects.